

21bis. Interfacce Android - Material Design II

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Overview

. components (widgets):

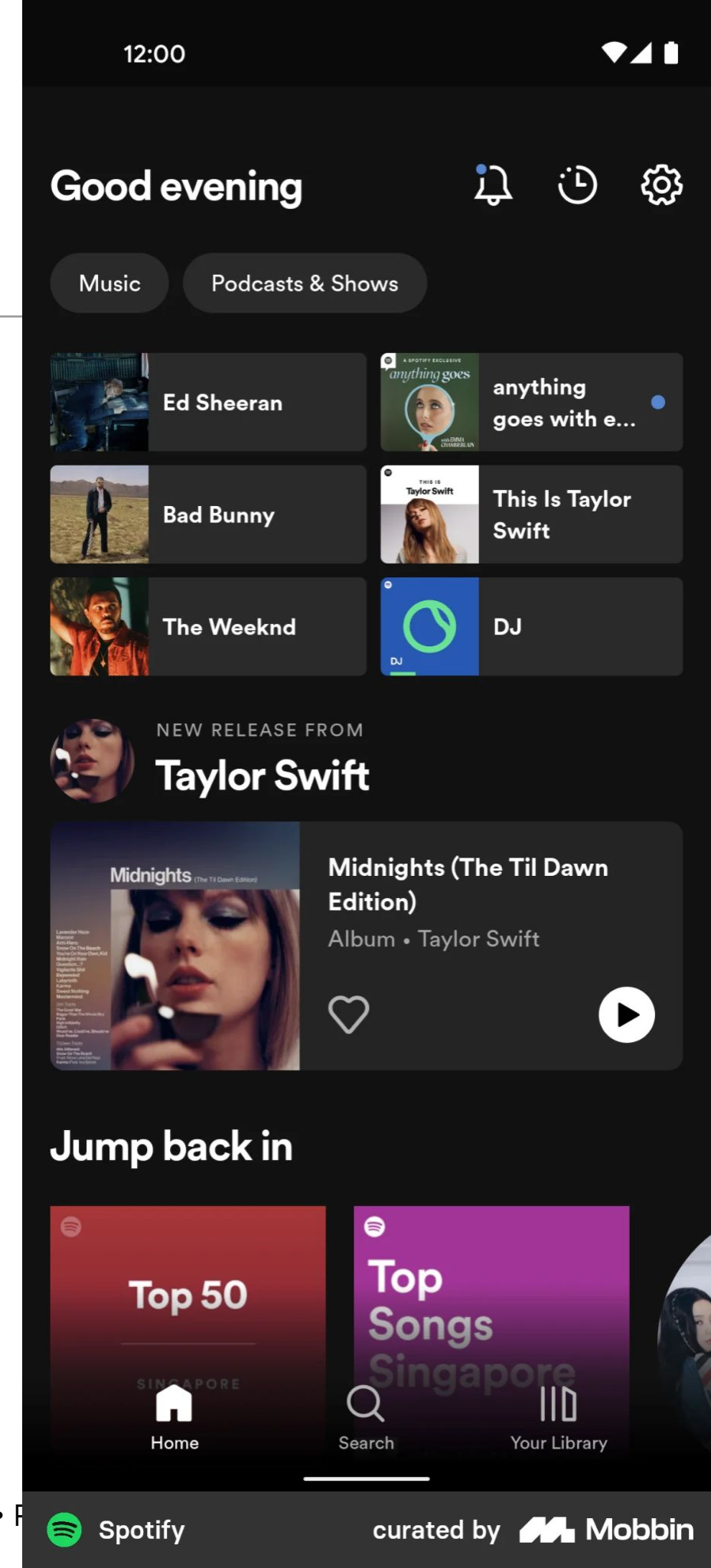
- cards <https://m3.material.io/components/cards/overview>
- lists <https://m3.material.io/components/lists/overview>
- buttons <https://m3.material.io/components/buttons/overview>
- floating action buttons <https://m3.material.io/components/floating-action-button/overview>
- bottom sheets <https://m3.material.io/components/bottom-sheets/overview>
- side sheets <https://m3.material.io/components/side-sheets/overview>
- tabs <https://m3.material.io/components/tabs/overview>
- bottom app bar <https://m3.material.io/components/bottom-app-bar/overview>
- top app bar <https://m3.material.io/components/top-app-bar/overview>
- navigation bar <https://m3.material.io/components/navigation-bar/overview>

. styles:

- transitions <https://m3.material.io/styles/motion/transitions/transition-patterns>

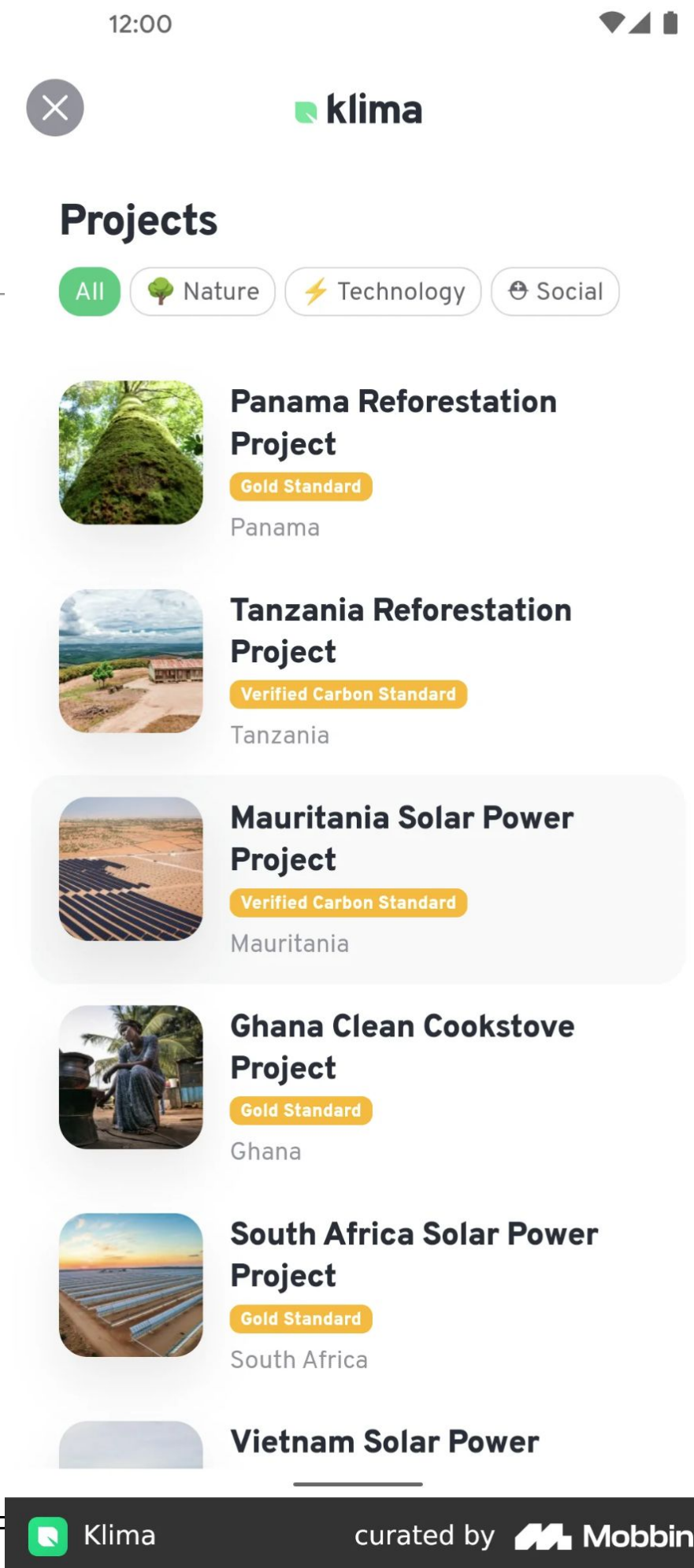
Cards

- use a card to display content and actions on a single topic
- entry point into deeper levels of detail or navigation
- cards can be displayed together in a grid, vertical list, or carousel
- content: headline, text, dividers, media, icons, images, buttons
- cards can be one large touch target triggering an expanded detail screen



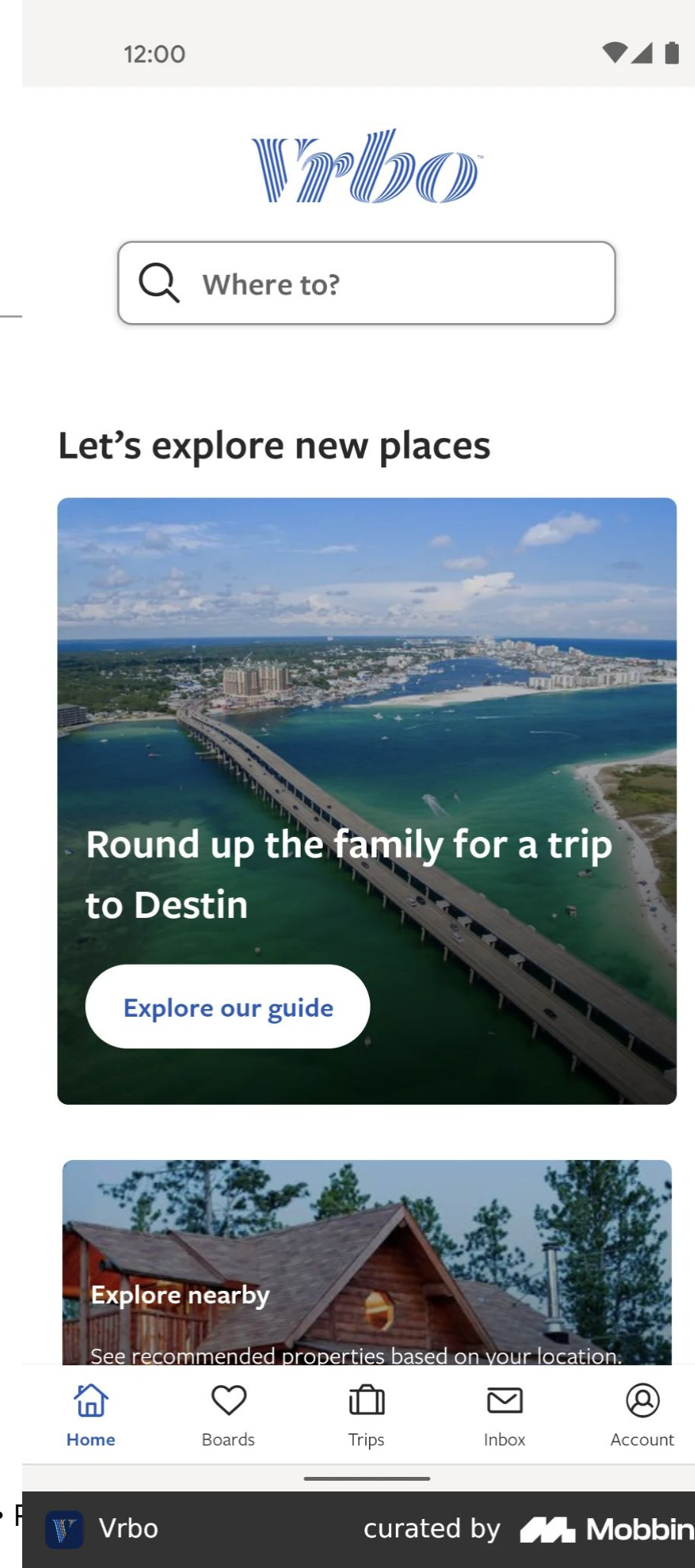
Lists

- continuous, vertical indexes of text and images
- help users find a specific item and act on it
- order in logical ways (like alphabetical or numerical)
- 3 sizes: one-line, two-line, and three-line
- keep items short and easy to scan
- show icons, text, and actions in a consistent format
- **align visuals to the leading edge**



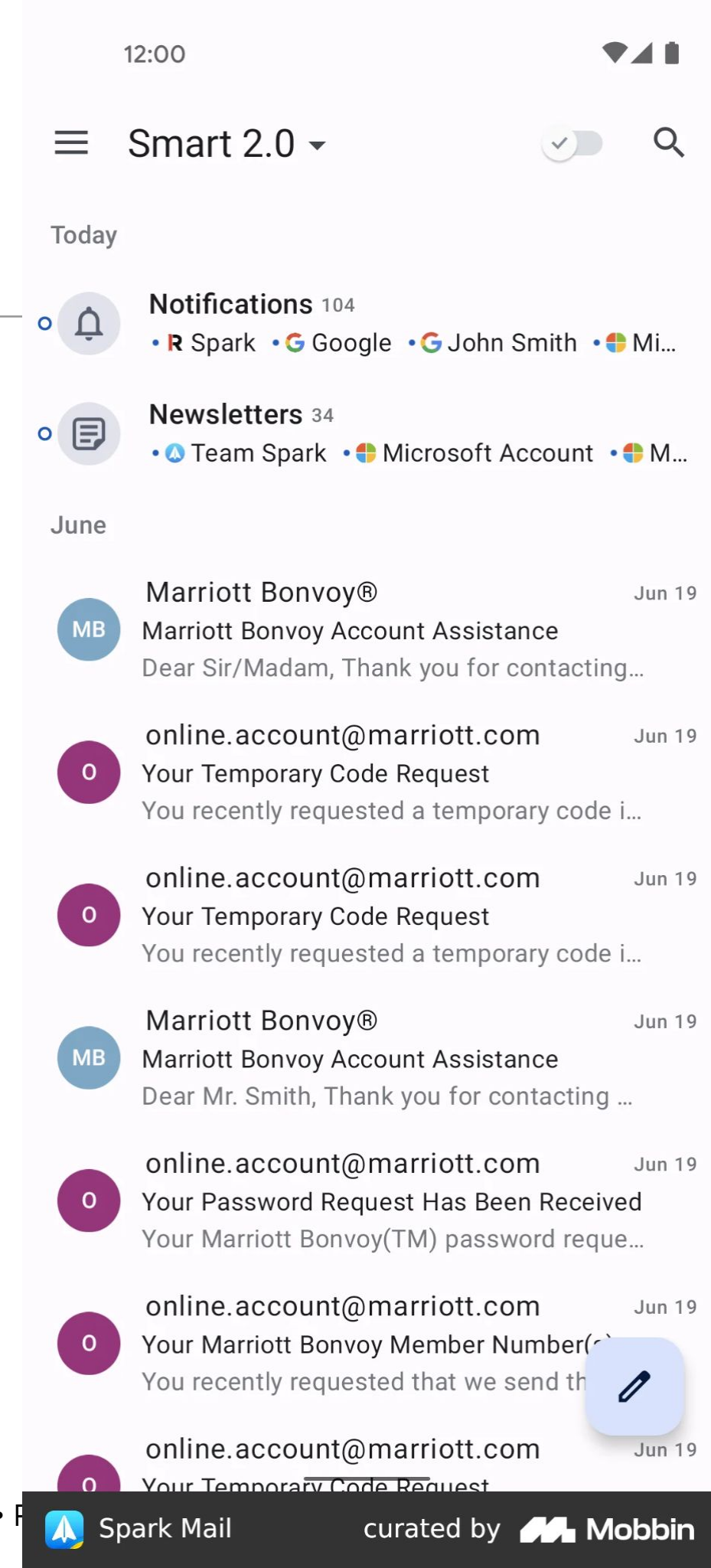
Buttons

- communicate actions that users can take
- containers have fully rounded corners and are wide enough to fit label text
- used in: dialogs, modal windows, forms, cards, toolbars
- too many buttons on a screen disrupts the visual hierarchy
- icons (optional) should be placed on the leading side of the button, before the label text



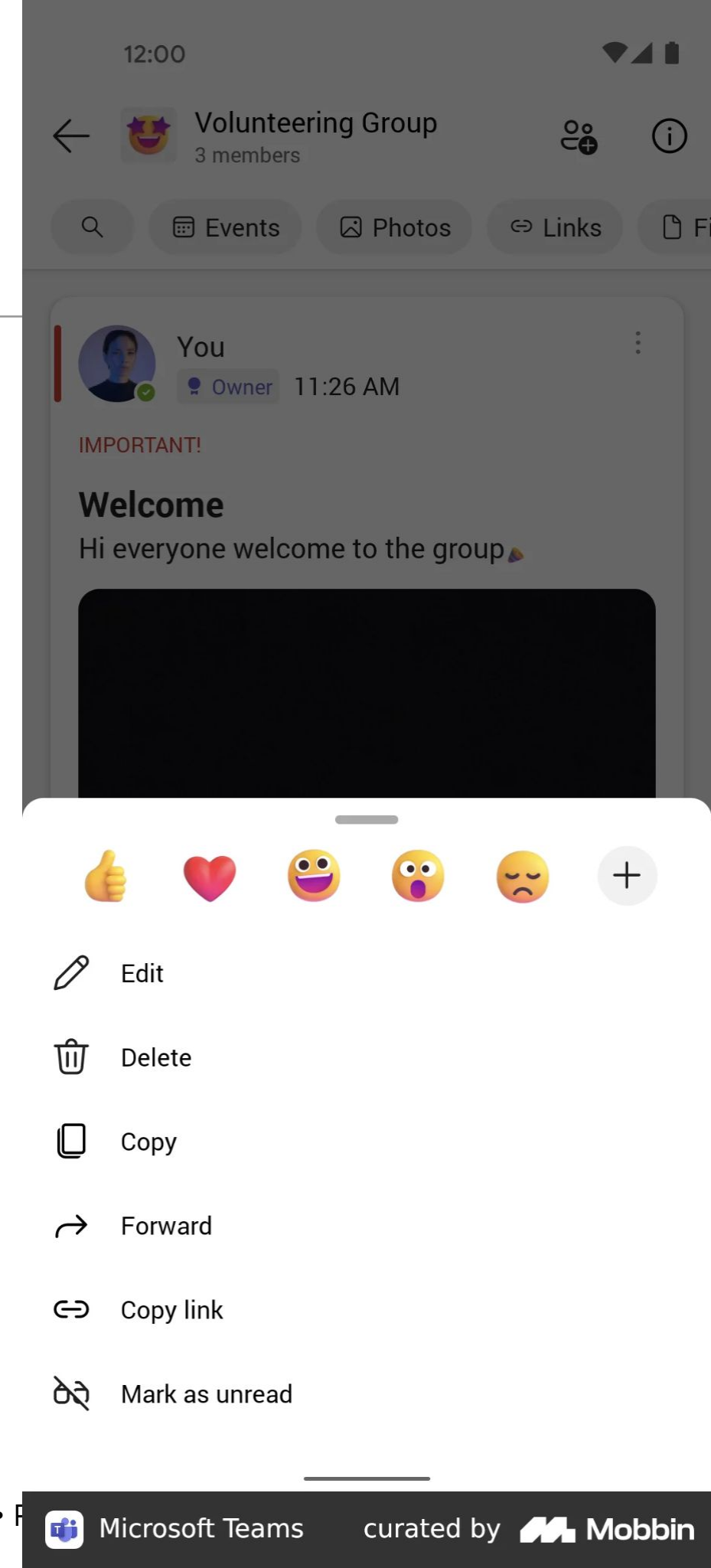
Floating Action Buttons (FAB)

- the most common or important action on a screen
- promotes a constructive action: create, favorite, share, start a process
- icon clear and understandable
- persists when content is scrolling
- aligned left, center, or right
- positioned above the navigation bar, or nested within the bar
- displayed in a rounded square container



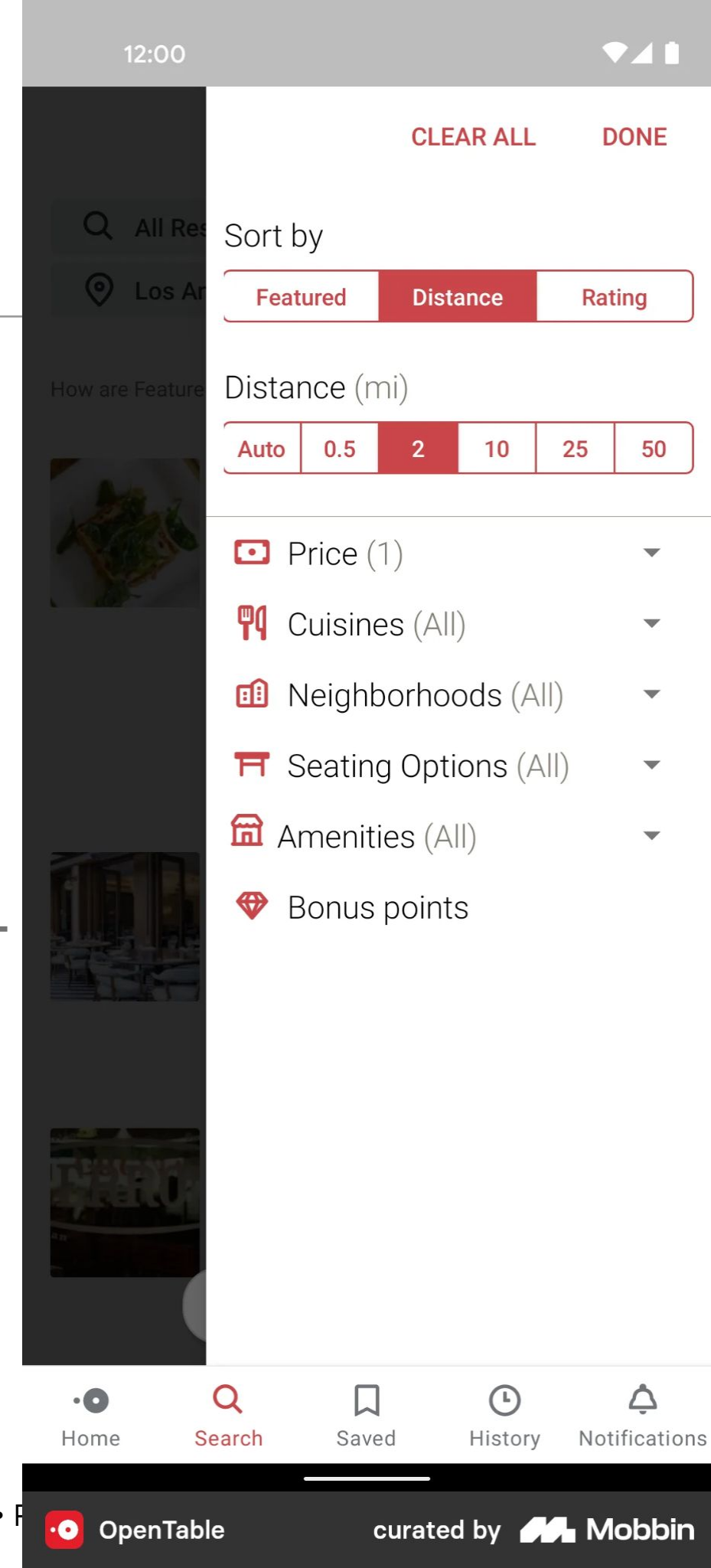
Bottom sheets

- display supplementary content and actions
- can contain menu items (in list or grid layouts), actions
- 2 types:
 - standard - displays content that complements the screen's primary content, such as an audio player in a music app
 - modal - an alternative to inline menus or simple dialogs on mobile; can then be pulled across the full screen and scrolled internally (tall)



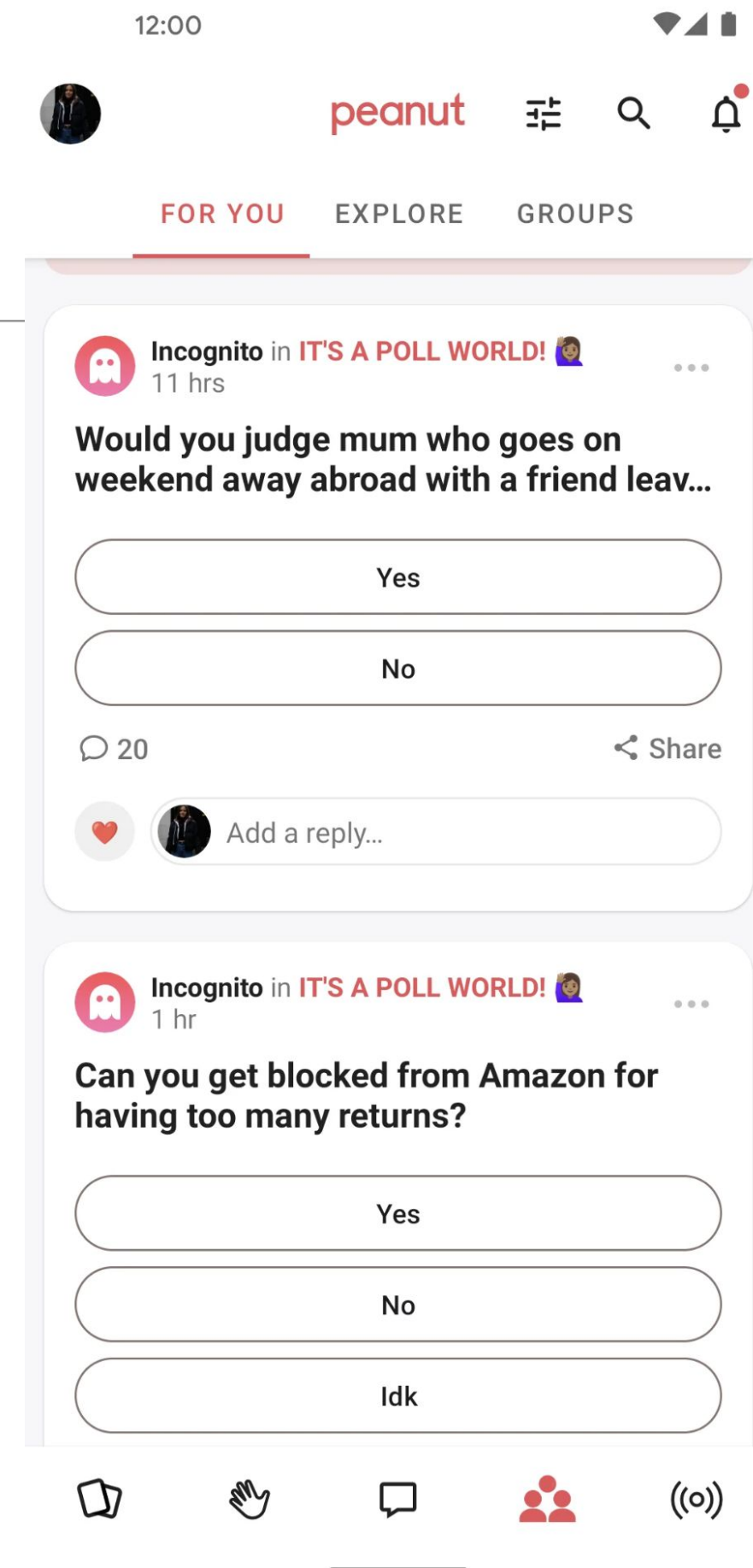
Side sheets

- show secondary content anchored to the side of the screen
- display a list of actions that affect the screen's primary content, such as filters
- predictive back allows user to swipe left or right on the side sheet
- 2 types:
 - standard - display content that complements the screen's primary content
 - modal - preferred in compact window sizes, like mobile; the same types of content as standard



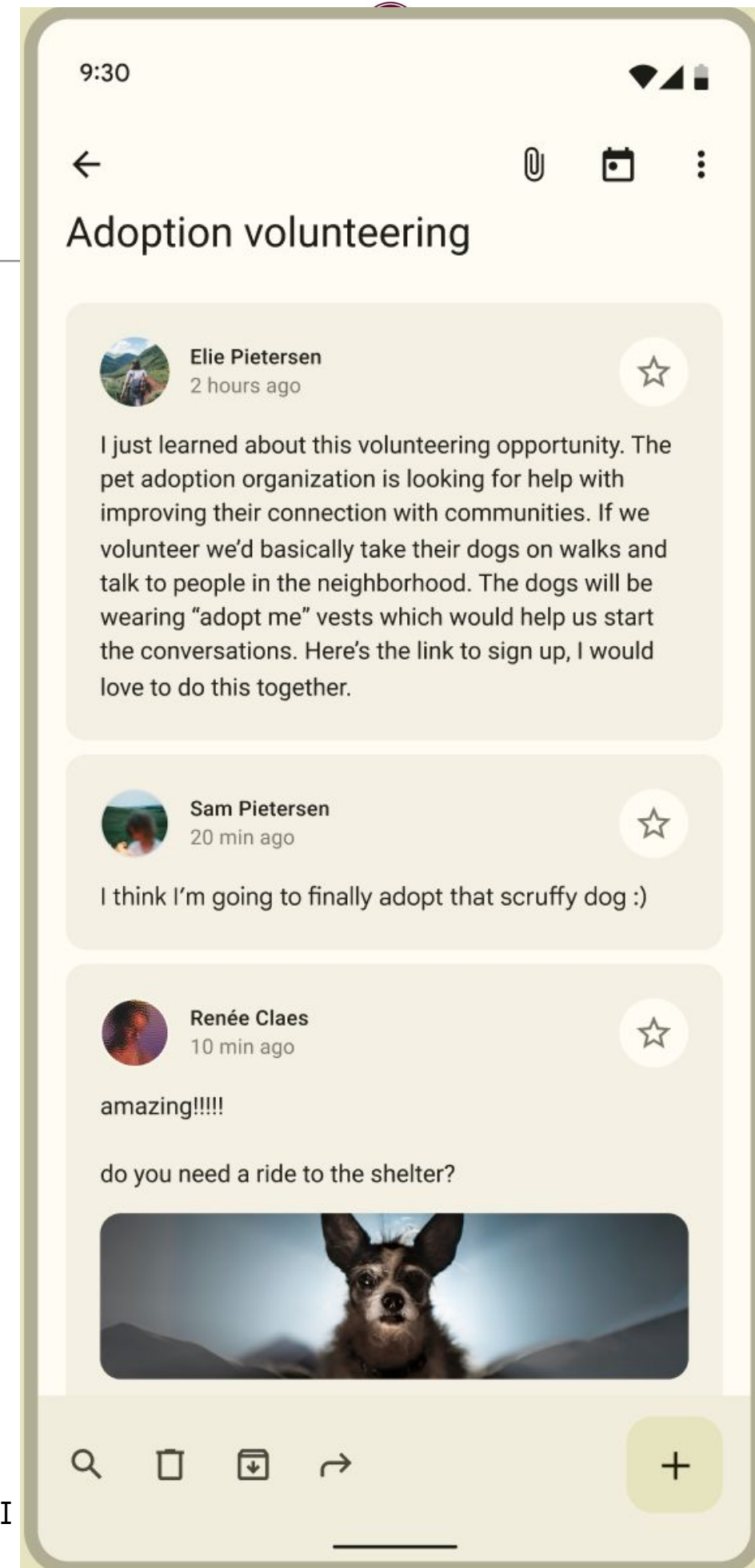
Tabs

- organize content across different screens and views
- group content into helpful categories
- placed at the top of the content pane under a top app bar
- label + icon (optional)
- apply an underline and color change to the active tab's text and icon
- to navigate between tabs, users can swipe left or right



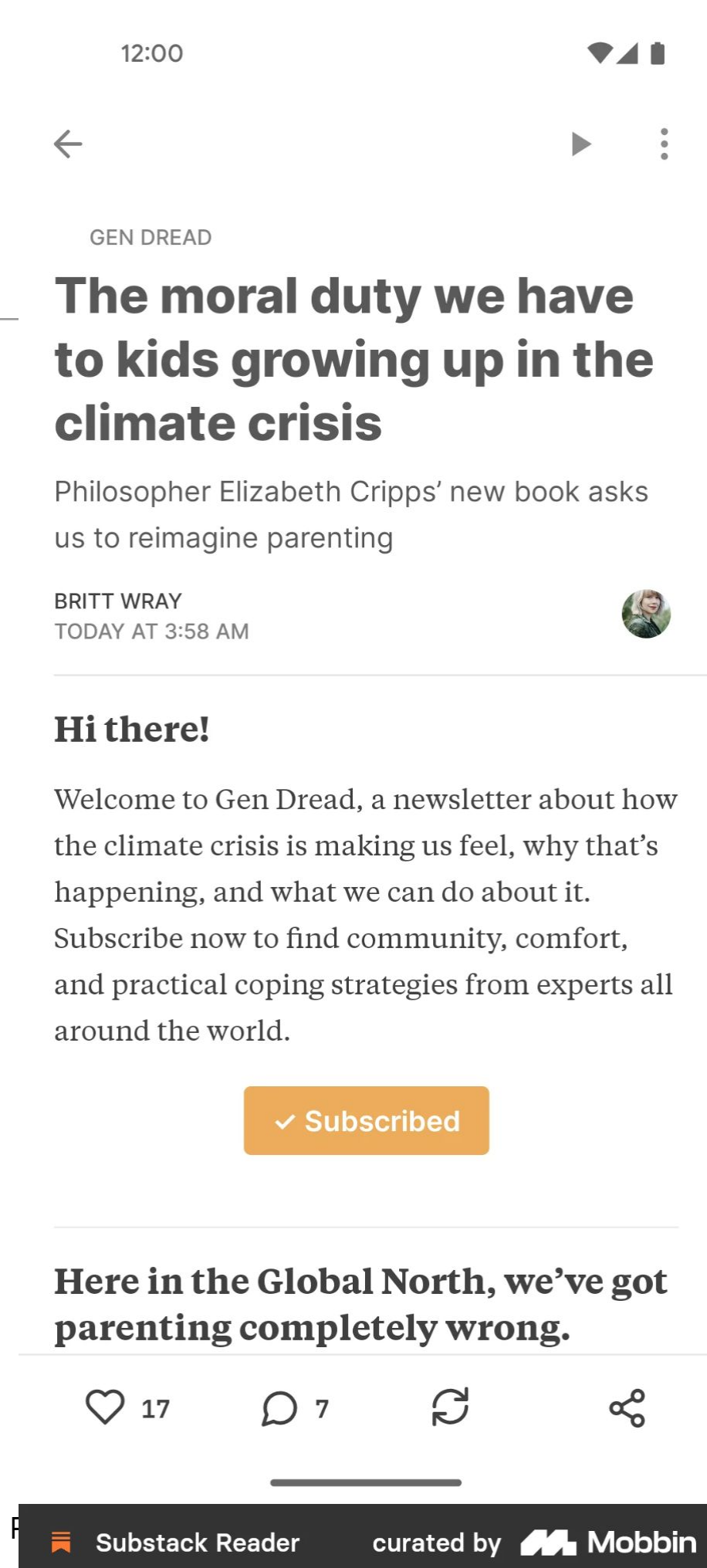
Bottom app bars

- display navigation and key actions
- up to four icon buttons and a FAB
- **shouldn't be used** for apps with a navigation bar or screens with one or no actions
- they aren't navigation bars
- when performing destructive action, should trigger a snackbar to confirm or undo the action
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Top app bar

- title and actions related to the current screen
- navigation, screen headlines, and actions
- contain an optional leading navigation icon, 2 forms:
 - a menu icon (**hamburger**)
 - a **back arrow**
- scrolling down hides the app bar
- scrolling up reveals the app bar



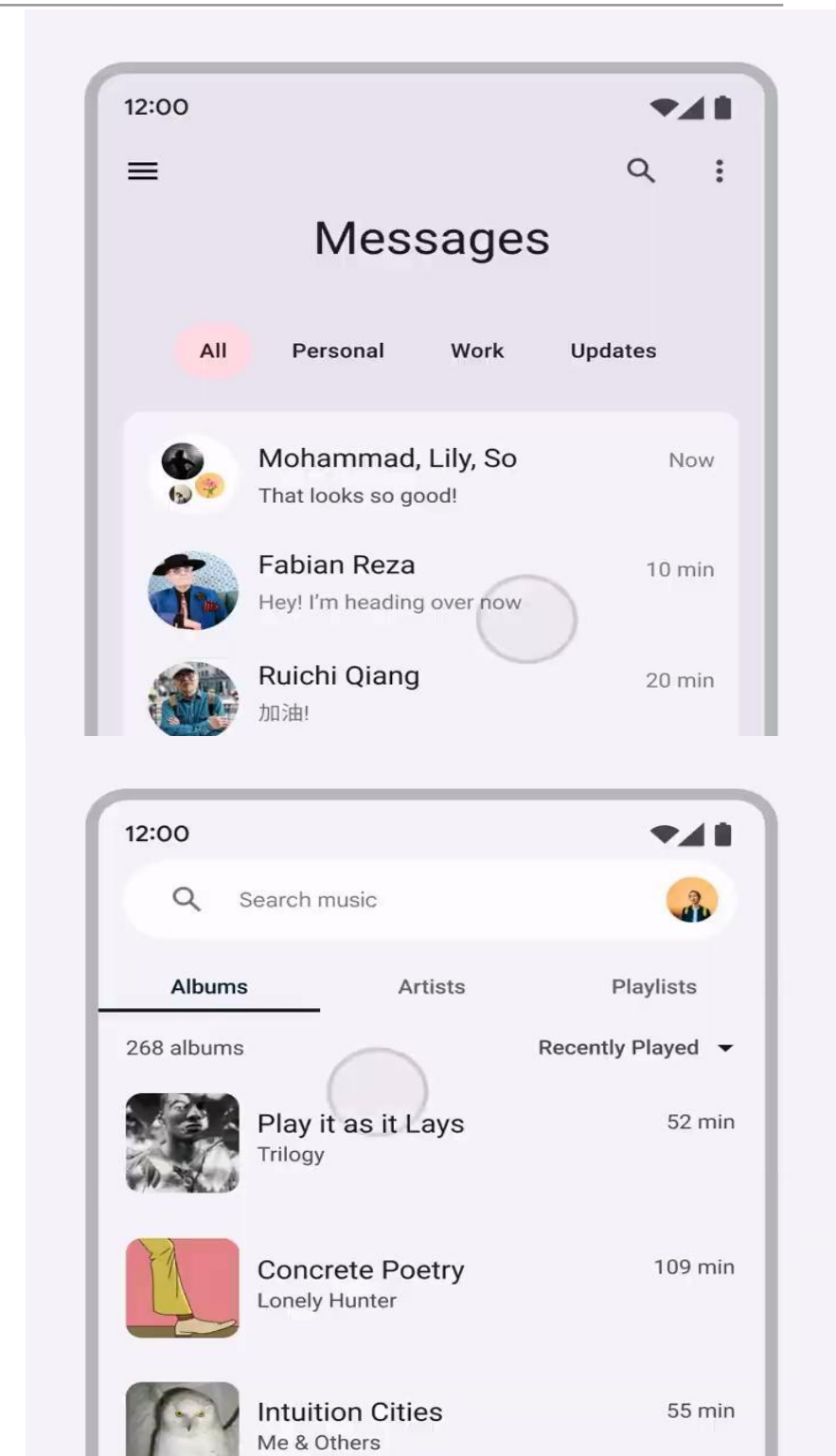
Navigation bars

- 3-5 destinations of equal importance
- used to be named bottom navigation
- icons and optional text labels
- use for top-level destinations that need to be accessible from anywhere in the app
- do not use for single tasks, such as viewing a single email, user preferences or settings



Transitions

- forward and backward
 - platform defaults for forward and backward navigation (e.g., hierarchy navigation)
- lateral
 - browse peer content that's part of the same set, like navigating between tabs in a media library



References

- Material Design 3
 - <https://m3.material.io/>
- Mobbin per Android
 - <https://mobbin.com/browse/android/apps>